The documentation and process conversion measures necessary to comply with this revision shall be completed by 27 August 1994

INCH-POUND

MIL-S-19500/453B
27 May 1994
SUPERSEDING
MIL-S-19500/453A(USAF)
21 July 1991

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, HIGH-FREQUENCY
TYPE 2N5109, JAN, JANTX, AND JANTXV

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for NPN silicon, VHF-UHF amplifier transistors. Three levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See 3.3 (TO-39).

1.3 Maximum ratings.

P_T 1/ 2/ T_A = +25°C	V_{CBO}	V_{CEO}	I_C	V_{EBO}	T_{STG}	T_J
W	V dc	V dc	A dc	V dc	°C	°C
1	40	20	0.4	3.0	-65 to +200	+200

1/ Derate at 5.71 mW/°C above T_A > +25°C.
2/ P_T = 2.9 W at T_C = +25°C, derate at 16.6 mW/°C above T_C > +25°C.

1.4 Primary electrical characteristics (common to all types).

Limits	h_{FE}	$V_{CE(SAT)}$	C_{obo}	$\|h_{FE}\|$	Power gain
	V_{CE} = 15 V dc I_C = 50 mA dc	I_C = 100 mA dc I_B = 10 mA dc	I_E = 0 100 kHz ≤ f ≤ 1 MHz V_{CB} = 28 V dc	V_{CE} = 15 V dc I_C = 50 mA f = 200 MHz	I_C = 50 mA dc f = 200 MHz Pin = -10 dBm V_{CE} = 15 V dc
		V dc	pF		dB
Min	40			6.0	11.0
Max	150	0.5	3.5	11.0	

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Electronics Supply Center, ATTN: DESC-ELDT, 1507 Wilmington Pike, Dayton, OH 45444-5765, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

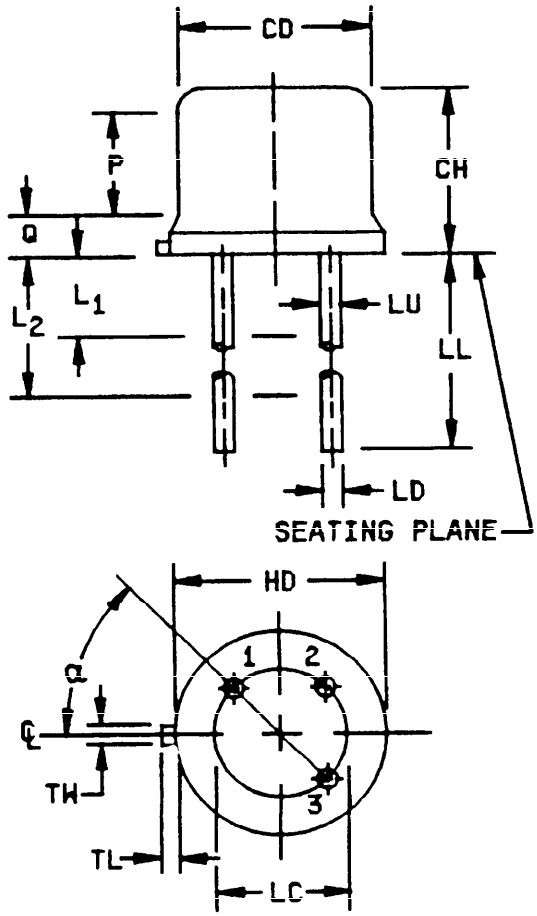
AMSC N/A

FSC 5961

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
LC	.190	.210	4.83	5.33	
CH	.240	.260	6.10	6.60	
LD	.016	.021	0.406	0.533	4
LU	.016	.019	0.406	0.483	4
HD	.335	.370	8.51	9.40	
CD	.305	.335	7.75	8.51	
TW	.028	.034	0.711	0.864	
TL	.029	.045	0.737	1.14	5
LL	.500	.750	12.70	19.05	4
L_1		.050		1.27	4
L_2	.250		6.35		4
P	.100		2.54		3
Q		.050		1.27	6
α	45° T. P.				

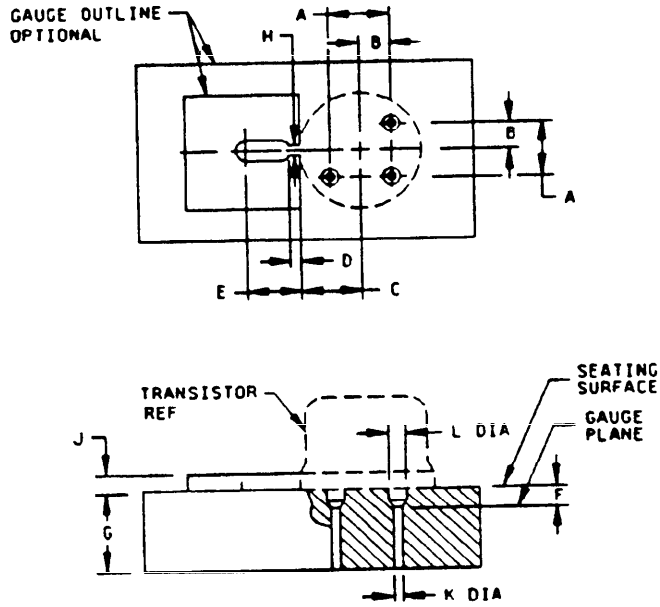


NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. This zone is controlled for automatic handling. The variation in actual diameter within this zone shall not exceed .010 inch (0.254 mm).
4. (Three leads) LU applies between l_1 and l_2. LD applies between l_2 and .5 inch (12.70 mm) from seating plane. Diameter is uncontrolled in l_1 and beyond .5 inch (12.70 mm) from seating plane.
5. Measured from maximum diameter of the actual device.
6. Details of outline in this zone optional.

FIGURE 1. Physical dimensions of transistor type 2N5109.

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Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.1409	.1419	3.579	3.604
B	.0702	.0712	1.783	1.809
C	.182	.199	4.62	5.05
D	.009	.011	0.23	0.28
E	.125 Nom		3.18 Nom	
F	.054	.055	1.37	1.40
G	.372	.378	9.45	9.60
H	.0350	.0355	0.889	0.902
J	.150 Nom		3.81 Nom	
K	.0325	.0335	0.826	0.851
L	.0595	.0605	1.511	1.537



NOTES:
1. Metric equivalents are given for general information only.
2. Dimensions are in inches.
3. The following gauging procedures shall be used: The use of a pin straightener prior to insertion in the gauge is permissible. The device being measured shall be inserted until its seating plane is .125 inch (3.18 mm) +.010 inch (0.254 mm) from the seating surface of the gauge. A spacer may be used to obtain the .125 inch (3.18 mm) distance from the gauge seat prior to force application. A force of 8 ±.5 oz. shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not be removed) the seating plane of the device shall be seated against the gauge.
4. The location of the tab locator, within the limits of dimension C, will be determined by the tab and flange dimension of the device being checked.

FIGURE 2. Gauge for lead and tab location.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Defense Printing Service Detachment Office, Bldg. 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500 and figures 1 and 2 herein (TO-39).

3.3.1 Lead finish. Lead finish shall be solderable in accordance with MIL-STD-750 and MIL-S-19500.

3.4 Marking. Marking shall be in accordance with MIL-S-19500. At the option of the manufacturer, the country of origin marking and the manufacturers identification may be omitted from the body of the transistor, but shall be retained on the initial container.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500, and as specified herein.

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4.3 Screening (JANTX and JANTXV levels only). Screening shall be in accordance with MIL-S-19500 (table II), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement
	JANTX and JANTXV levels
9	Not applicable
11	I_{CEO1} and h_{FE1}
12	See 4.3.1
13	ΔI_{CEO1} = 100% of initial value or 2 µA dc, whichever is greater. Δh_{FE1} = 20% change from initial reading; Subgroup 2 of table I herein.

4.3.1 Power burn-in. Power burn-in conditions are as follows:
T_A = Room ambient as defined in the general requirements of MIL-STD-750, section 4.5;
V_{CB} = 15 V dc; P_T = 1.0 W.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein.

4.4.2 Group B inspection (JANTX and JANTXV). Group B inspection shall be conducted in accordance with conditions specified for the subgroup testing in table IVb (JANTX and JANTXV) of MIL-S-19500 and as follows. Electrical measurements (end-points) shall be in accordance with the applicable steps and footnotes of table II herein.

Subgroup	Method	Condition
B3	1027	V_{CB} = 15 V dc; P_T = 1.0 W; T_A = +25°C ±5°C. No heat sink or forced air cooling on the device shall be permitted.
B3	2037	Test condition A.
B5	---	Not applicable.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500 and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

Subgroup	Method	Condition
C2	2036	Tension: Test condition E.
C6	1037	V_{CB} = 15 V dc at T_A = +25°C ±5°C; P_T = 1.0 W. No heat sink or forced air cooling on the device shall be permitted.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Breakdown voltage, collector to base	3001	Bias condition D; I_C = 100 μA dc	$V_{(BR)CBO}$	40		V dc
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 5 mA dc	$V_{(BR)CEO}$	20		V dc
Breakdown voltage, emitter to base	3026	Bias condition D; I_C = 100 μA dc	$V_{(BR)EBO}$	3.0		V dc
Breakdown voltage, collector to emitter	3011	Bias condition B; I_C = 5 mA dc; R_2 = 100Ω	$V_{(BR)CER}$	40		V dc
Collector to emitter cutoff current	3041	Bias condition D; V_{CE} = 15 V dc	I_{CEO1}		20	μA dc
Forward-current transfer ratio	3076	V_{CE} = 15 V dc; I_C = 50 mA dc; Pulsed (see 4.5.1)	h_{FE1}	40	150	
Collector to emitter saturated voltage	3071	I_C = 100 mA dc; I_B = 10 mA dc; Pulsed (see 4.5.1)	$V_{CE(sat)1}$		0.5	V dc
Subgroup 3						
High temperature operation:		T_C = +175°C				
Collector to emitter cutoff current	3041	Bias condition D; V_{CE} = 15 V dc	I_{CEO2}		5.0	mA dc
Low temperature operation:		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 50 mA dc; Pulsed (see 4.5.1)	h_{FE2}	15		
Subgroup 4						
Open circuit output capacitance	3236	V_{CB} = 28 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		3.5	pF

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4 - Continued.						
Magnitude of common-emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 15 V dc; f = 200 MHz	$\|h_{fe}\|$			
		I_C = 25 mA dc;		5.0	10	
		I_C = 50 mA dc;		6.0	11.0	
		I_C = 100 mA dc		5.0	10.5	
Power gain (narrow band) current		V_{CC} = 15 V dc; I_C = 50 mA dc; f = 200 MHz; Pin = -10 dB; (See figure 3)	G_{PE}	11		dB
Cross modulation		V_{CC} = 15 V dc; I_C = 50 mA dc; 54 dBm V output; (See figure 4)	cm		-57	dB
Noise figure		V_{CC} = 15 V dc; I_C = 10 mA dc; f = 200 MHz; (See figure 3)	NF		3.5	dB
Voltage gain (wideband)		V_{CC} = 15 V dc; I_C = 50 mA dc; f = 50 to 216 MHz; (See figure 4)	G_{VE}	11		dB
Subgroups 5, 6, and 7						
Not applicable						

1/ For sampling plan, see MIL-S-19500.

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TABLE II. Groups B and C electrical end-point inspection measurements. 1/ 2/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Collector to emitter cutoff current	3041	Bias condition D; V_{CE} = 15 V dc	I_{CEO1}		20.0	µA dc
2.	Collector to emitter cutoff current	3041	Bias condition D; V_{CE} = 15 V dc	I_{CEO2}		40.0	µA dc
3.	Breakdown voltage, emitter to base	3026	Bias condition D; I_E = 100 µA dc	BV_{EBO}	3.0		

1/ The electrical measurements for table IVb (JAN, JANTX, and JANTXV) of MIL-S-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1 and 3.

b. Subgroup 3, see table II herein, steps 2 and 3.

c. Subgroup 6, see table II herein, steps 2 and 3.

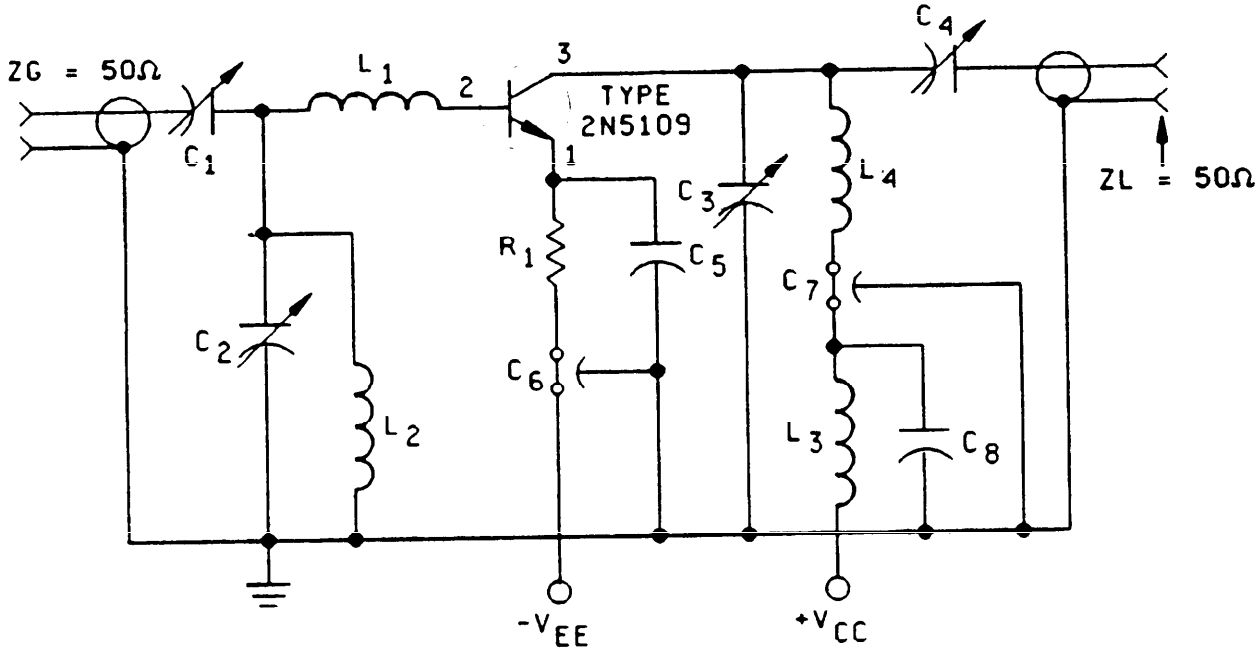
2/ The electrical measurements for table V of MIL-S-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1 and 3.

b. Subgroup 3, see table II herein, steps 1 and 3.

c. Subgroup 6, see table II herein, steps 2 and 3.

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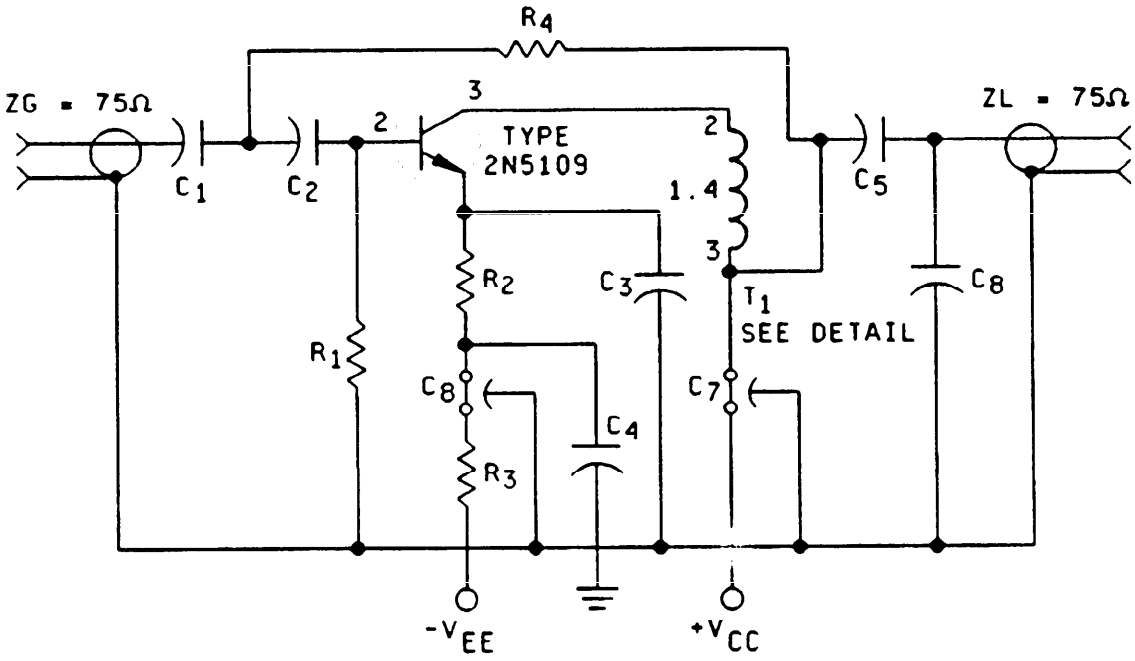


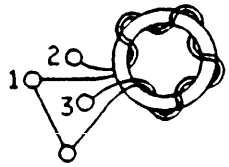
C_1, C_2, C_3: 1.0 to 30 pF.
C_4: 1.0 to 20 pF.
C_5: 10,000 pF.
C_6, C_7: 1,000 pF.
C_8: .01 µF.

L_1: 4.5 turns, number 22 wire, .187 inch I. D.
L_4: 3.5 turns, number 22 wire, .187 inch I. D.
L_2, L_3: 0.82 µH RFC.
R_1: 240Ω, 2 watts.

FIGURE 3. RF amplifier for power gain and noise figure tests.

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DETAIL OF TRANSFORMER T_1

C_1, C_2, C_3, C_5: 0.002 µF.
C_4: 0.03 µF.
C_6, C_7: 1500 pf.
C_8: 18 pf.
R_1: 4.7 kΩ, 0.25 W.
R_2: 6.8Ω, 0.5 W.
R_3: 330Ω, 1 W.
R_4: 200Ω, 0.25 W.
T_1: 4 turns number 30 wire.

Bifilar wound on "Indiana General".
Core number CF-102-Q1, or equivalent.

FIGURE 4. RF amplifier for voltage gain test.

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4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1).

c. Lead finish (see 3.3.1).

d. Product assurance level and type designator.

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue because of the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Air Force - 17

Review activities:
Air Force - 19, 85, 99

Preparing activity:
DLA - ES

(Project 5961-F091)

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/453B	2. DOCUMENT DATE (YYMMDD) 94/05/27

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, HIGH-FREQUENCY TYPE 2N5109, JAN, JANTX, AND JANTXV

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE (Include Area Code) (1) Commercial 513-296-6048	(2) AUTOVON 986-6048
c. ADDRESS (Include Zip Code) Defense Electronics Supply Center ATTN: DESC-ELDT Dayton, OH 45444-5765	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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